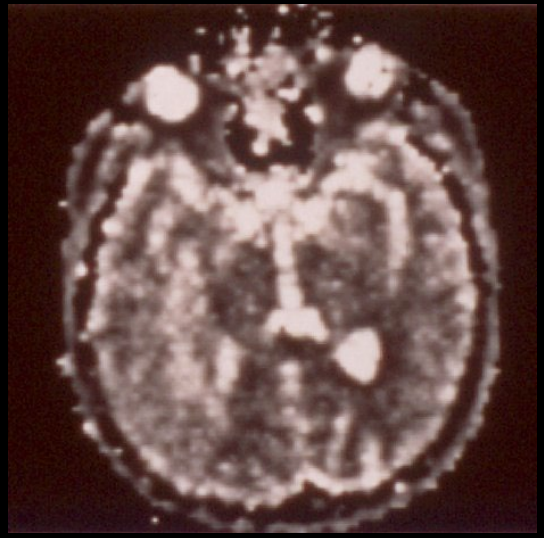
Introduction in Magnetic Resonance Imaging



Andreas Hock

MRI (Magnetic Resonance Imaging)

Historical overview

Fundamental Physics Behind MRI

1. Nuclear Magnetic Resonance (NMR) – Felix Bloch ⭐ and **Edward Purcell** ⭐ (1946)

- Discovered NMR independently.
- Shared the 1952 Nobel Prize in Physics.

2. Bloch Equations – Felix Bloch ⭐

- Mathematical description of NMR signal behavior in a magnetic field.

3. Spin Echo Technique – Erwin Hahn (1950)

- Foundation for eliminating signal dephasing and improving image quality.

Development of MRI Technology

4. Proposal of MRI for Medical Imaging – Raymond Damadian (1971)

- Showed that cancerous and normal tissues have different NMR relaxation times.
- Controversially not awarded the Nobel.

5. First MRI Image (Ex Vivo) – Paul Lauterbur ⭐ (1973)

- Introduced the concept of spatial encoding using magnetic field gradients.
- Shared the 2003 Nobel Prize in Physiology or Medicine.

6. Phase and Frequency Encoding – Paul Lauterbur ⭐

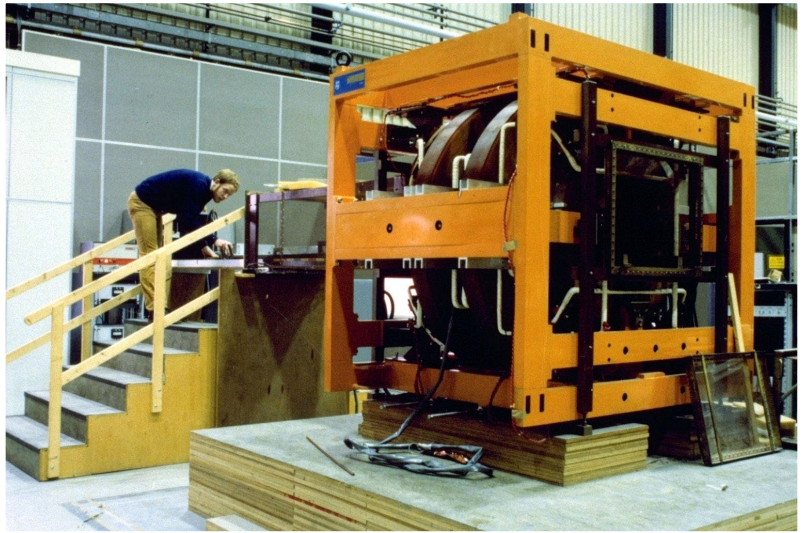
- Critical for turning NMR data into 2D and 3D images.

7. Fourier Transform Imaging – Peter Mansfield ⭐

- Developed mathematical techniques for rapid image reconstruction.
- Shared the 2003 Nobel Prize in Physiology or Medicine.

8. Echo Planar Imaging (EPI) – Peter Mansfield ⭐

- Enabled real-time imaging and fMRI (functional MRI).



Prototype MR-System, 1981 (0.15 Tesla)

⭐ Nobel prize

MRI (Magnetic Resonance Imaging)

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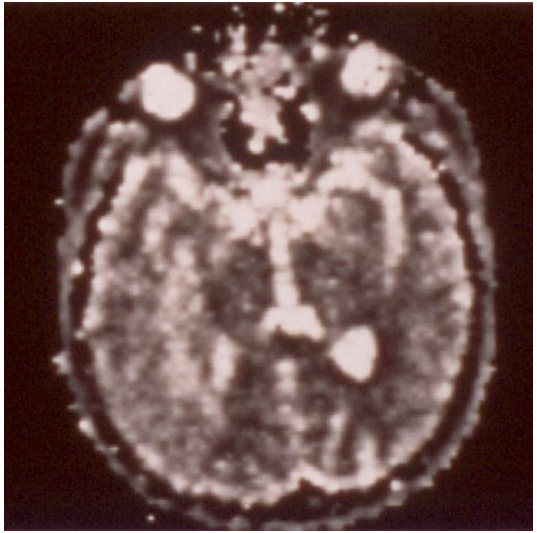
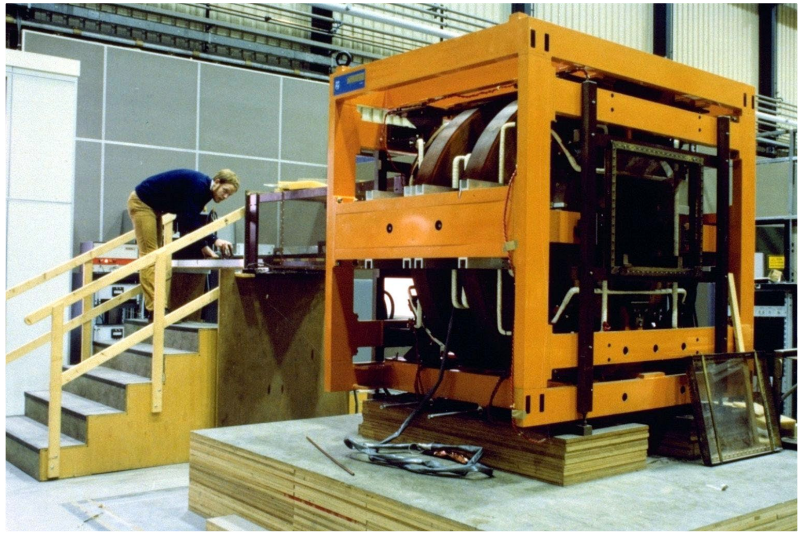


MR image 1981 (0.15 Tesla)



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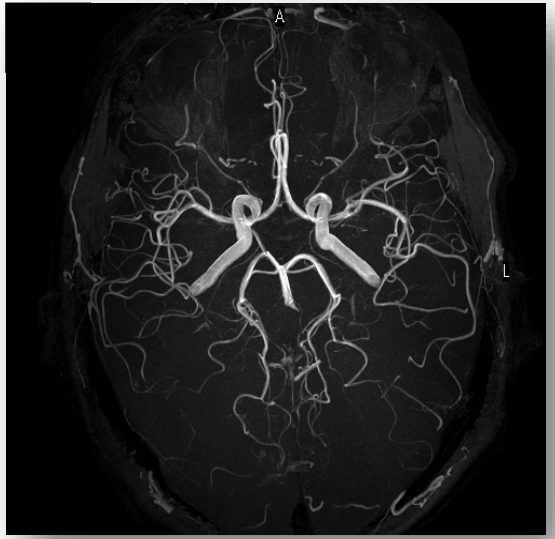


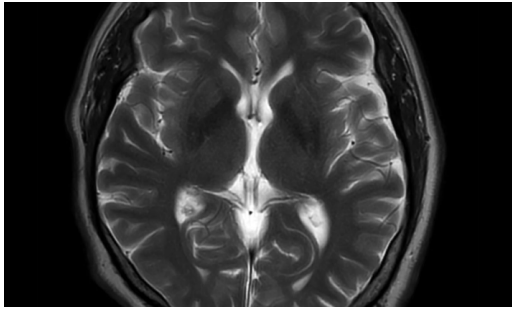
2048 matrix (4 megapixels)

3 Tesla
36 layers
5 mm layer thickness
Recording
2:42 Min.

MR-System 2019 (1.5 or 3.0 Tesla)

MRI

Advantages

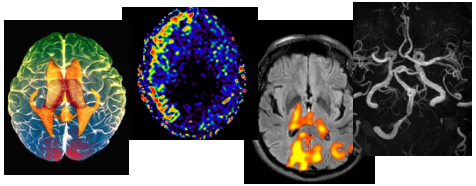


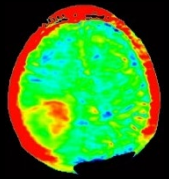
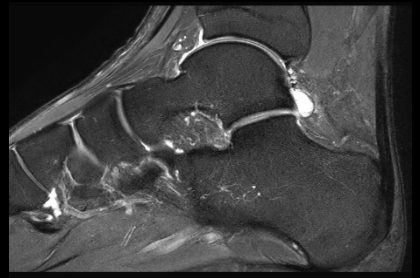
- High soft tissue contrasts
- Entire body can be measured
- No ionizing radiation (non-invasive)
- Multi-Planar Imaging (Arbitrary layer orientations)

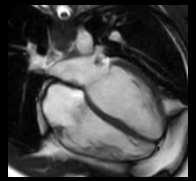
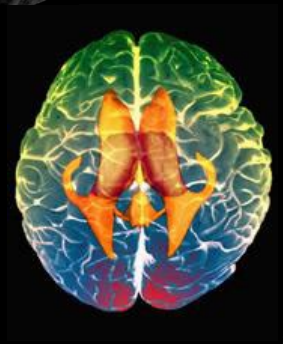
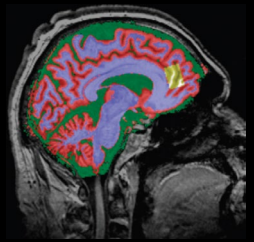
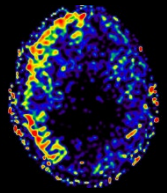
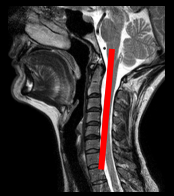
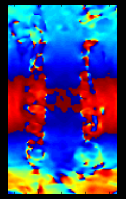
- Software controlled

- Many possible applications

(anatomy, diffusion, function, perfusion, chemical information, temperature, ...)



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